

$$A = \begin{bmatrix} 9 & 15 \\ 15 & 50 \end{bmatrix} \Rightarrow A \text{ is symmetric}$$

$\rightarrow A$ can be written as LL^T

$$\text{Let } L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}$$

$$LL^T = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 15 & 50 \end{bmatrix}$$

$$l_{11}^2 = 9 \quad - (1)$$

$$l_{11}l_{21} = 15 \quad - (2)$$

$$l_{21}^2 + l_{22}^2 = 50 \quad - (3)$$

Solving (1), (2) & (3)

$$|l_{22}| = 5$$